

ServiceNow Certified Application Developer

SmartInternz

Project Title

**Automated Network Request
Management in ServiceNow**

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Project overview

The **Automated Network Request Management** project in ServiceNow is designed to make network service requests faster and easier to manage. Users can submit network-related requests through the **Service Catalog** using a self-service portal.

The system automatically **routes requests, manages approvals, assigns tasks, and tracks their progress**. By reducing manual work and automating repetitive processes, the solution helps IT teams complete requests quickly and efficiently.

This improves **turnaround time, SLA compliance, request visibility, and overall user experience**.

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1. Introduction

Organizations depend on network services for their daily operations. Employees may need different network services such as new connections, device connectivity, relocation, or other network-related support.

Handling these requests manually can take more time and may lead to errors or delays. It can also be difficult for users and IT teams to track the current status of a request.

The **Automated Network Request Management in ServiceNow** project provides a simple solution to manage these requests in an organized and automated way.

Users can submit their network requests through the **Service Catalog**. Based on the submitted request, ServiceNow can collect the required information, create records, send them for approval, assign the required tasks, and send notifications.

This reduces manual work and helps the IT team process requests faster and more efficiently.

2. Project Objective

The main objective of this project is to **automate the process of handling network service requests in ServiceNow**.

The system allows users to submit requests through a self-service catalog. The required information is collected through a structured form.

After the request is submitted, the system can automatically:

- Capture the request details
- Create a network request record
- Send the request for approval
- Process the approval result
- Send email notifications
- Update the request record
- Track the request status

The main goal is to **reduce manual effort, improve processing speed, and provide better visibility of network requests.**

3. Key Features

The major features of the project are: **3.1**

Service Catalog

A dedicated **Network Request** catalog item is created so users can submit their network requirements.

3.2 Dynamic Form

The form contains different fields to collect the information required to process the request.

3.3 Automated Approvals

Requests can be automatically sent for approval based on the configured process.

3.4 Flow Designer

Flow Designer is used to automate the complete request process.

3.5 Record Creation

When a user submits a request, the required information can be used to create a record in the Network Database table.

3.6 Email Notifications

The system can send email notifications to the required users or teams.

3.7 Request Tracking

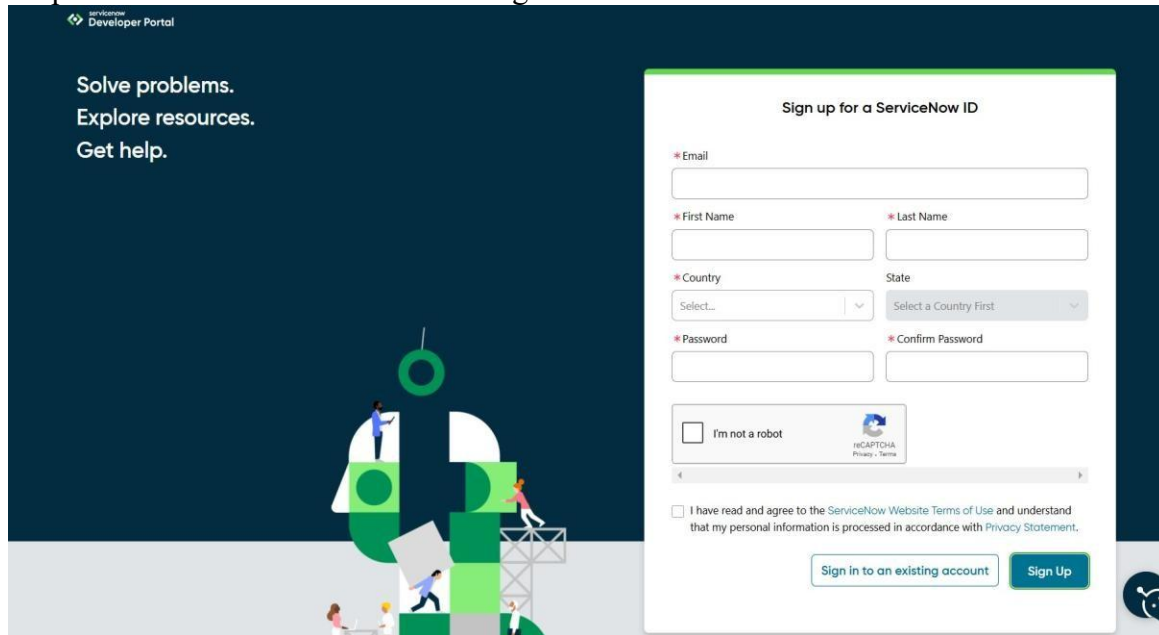
The request information and its progress can be tracked through ServiceNow.

These features help reduce repetitive manual activities and make the request process more organized.

4. ServiceNow Developer Setup:

Create a Developer Account

1. Go to ServiceNow Developer Portal(<https://developer.servicenow.com/dev.do>). Sign up for a free developer account and fill out the following details.

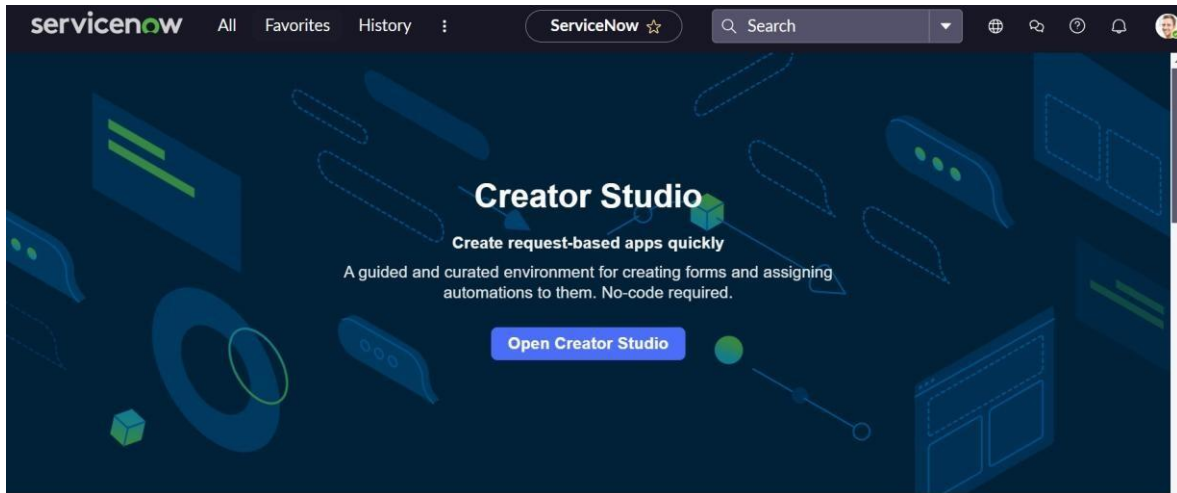
The image shows the ServiceNow Developer Portal sign-up page. On the left, there's a dark blue background with the text "Solve problems. Explore resources. Get help." and an illustration of people working on a large green and white puzzle. On the right, there's a white sign-up form titled "Sign up for a ServiceNow ID". The form includes fields for Email, First Name, Last Name, Country (a dropdown menu), State (a dropdown menu), Password, and Confirm Password. There is a checkbox for "I'm not a robot" with a reCAPTCHA logo. At the bottom, there's a checkbox for "I have read and agree to the ServiceNow Website Terms of Use and understand that my personal information is processed in accordance with Privacy Statement." and two buttons: "Sign in to an existing account" and "Sign Up".

2. After signing up, you will get a verification mail to your provided email id. After the verification your ServiceNow Developer Portal Home Page will appear.
3. Now click on start building it will take you to the section where you can **request a Personal Developer Instance (PDI)** or start using **App Engine Studio** and other tools.
4. **Profile Icon (Top Right Corner)** → Manage your account, request instances, and check your developer profile.

5. Project Implementation in ServiceNow:

After the instance building is completed, the page will be directed to your creator studio.

Creator Studio in ServiceNow provides a **guided, no-code environment** to build applications quickly. It is especially useful for creating **request-based applications** by defining forms, setting up tables, and automating workflows.



a. Service Catalog Creation

i. Creation of Service Catalog

1. Navigate to Application navigator
2. Click on All >> search for Service Catalog
3. Under Service Catalog>> Maintain items
4. Click on New
5. Fill the details >> Name– Network Request
6. Select Catalog>> Service Catalog
7. Select Category>> Network
8. Fill the Short Description as Network request Management 9. Click on Save.

< ≡ Catalog Item
 Network Request

Copy Try It Update Edit in Catalog Builder Delete

Catalog items are goods or services available to order from the service catalog. Items can be anything from hardware, like tablets and phones, to software applications, to furniture and office supplies.

- Enter a Name and Short description to display for the item.
- Enter a Price, approvals, variables, and other information as needed.

Name Network Request

Application Global

Catalogs Service Catalog

Category Networks and Connectivity

State -- None --

Checked out -- None --

Owner System Administrator

Active ☒

Fulfillment automation level Unspecified

ii. Variables Configuration

Open the catalog item just created.

Scroll down to the **Variables** related list and click **New** to create form fields.

1. Select Variables type as Single, Multi line text, reference, choices etc as per requirement
2. Catalog item– Network Request
3. Order–100,200,300,,,
4. Question– provide the variable label
5. Name–provide the variables name(used for scripting)
6. Tooltip– this will appear when cursor overed on the field
7. Example text – this will suggest what we need to enter on the field.
8. Mandatory, Read-Only– need to configure on demand.
9. Auto populate– need to select dependent variable, apply dot walking to get selected value.
10. Click on Save or Submit.

Variables (10)	Variable Sets (1)	Catalog UI Policies (1)	Catalog Client Scripts	Available For	Not Available For	Categories (1)	Catalogs (1)	Catalog Data Lookup Definitions	Related Articles	Related Catalog Items
Assigned Topics										
Order Search ⊙ ⚡ — Actions on selected rows... New										
Catalog Item = Network Request										
☐	☐	Type	Question				Order			
		Container Start	Service Details				200			
		Multiple Choice	Is this a New connection or Relocation?				300			
		Single Line Text	If this is a relocation, Please provide ...				310			
		Single Line Text	If this is a relocation, Please provide ...				320			
		Container Start	Location & Devices Type				400			
		Single Line Text	Please provide address here				410			
		Multiple Choice	Types of Devices				420			
		Single Line Text	Provide device details here				430			
		Container Start	Additional information				500			
		Single Line Text	If anything else, please specify				510			
1 to 10 of 10 >										

iii. Variables Types

1. Is this a New connection or Relocation? >> **Choice** >> **New/ Relocation/None**
2. If this is a relocation, Please provide your relocated address here>>**String**
3. Types of devices>> **Choice**>> **Laptop/Mobiles/Others**
4. Please provide address here>>**String**
5. Provide device details here>> **String**
6. If anything else, please specify>> **String**

iv. Variable Set Configuration

To enhance form usability:

- Navigate to the **Variable Sets** (optional).
- Follow the same procedure as we used for Variables Creation, for the variable set as well.
- Apply variable sets to the catalog item.

Variables (10) **Variable Sets (1)** Catalog UI Policies (1) Catalog Client Scripts Available For Not Available For Categories (1) Catalogs (1) Catalog Data Lookup Definitions Related Articles Related Catalog Items

Assigned Topics

≡ ▾ Order ▾ Search ⓘ ⓘ — Actions on selected rows... ▾ New Edit...

Catalog Item = Network Request

☐ 🔍 **Variable set** Order ▴

Requester information 100

◀◀ ◀ 1 to 1 of 1 ▶ ▶▶

Variables Types

1. Opened on behalf of >> Reference>> reference to user table
2. Email Id >> Single line text >> Auto populate by Opened on behalf of variable.
3. User name >>Single line text >> Auto populate by Opened on behalf of variable.
4. Phone Number >>Single line text >> Auto populate by Opened on behalf of variable. 5.

Proof of Document >> Attachment

Variables (5)				
Catalog UI Policies				
Catalog Client Scripts				
Included In (1)				
Catalog Data Lookup Definitions				
Variable set = Requester information				
Name	Type	Question	Order	
email_id	Single Line Text	Email Id		
proof_of_document	Attachment	Proof of Document		
opened_on_behalf_of	Reference	Opened on behalf of		
username	Single Line Text	User name		
phonenumber	Single Line Text	Phone Number		

v. Catalog UI Policy Configuration

Scenario: If user selects types of devices is **Others**, then Please specify field should populate.

Procedure:

1. Navigate to catalog items
2. Open Network Request item
3. In related list, we have Catalog UI policy
4. Click on New button to configure New UI policy
5. Select Applies to as Catalog item
6. Select catalog item as Network Request
7. Provide short description, if required
8. Apply condition>> **types of devices is others**

9. Click on save, after saving the form will get UI policy actions in the related list

10. Click on New button to configure new UI Policy action, and Select the variable which we want to display on condition

11. Make Visible True as per our requirement

12. Update the UI Policy and Test the same on Catalog form

Variables (10)	Variable Sets (1)	Catalog UI Policies (2)	Catalog Client Scripts	Available For	Not Available For	Categories (1)	Catalogs (1)	Catalog Data Lookup Definitions	Related Articles	Related Catalog Items
Assigned Topics										
≡ ▼ Order ▼ Search 🔍 🔗 — Actions on selected rows... ▼ New										
Catalog item = Network Request										
☐	☐	Short description	Variable set	Conditions	Reverse if false	On load	Inherit	Updated	Order ▲	
		Types of devices is others	(empty)		true	true	false	2025-09-17 07:51:06	100	
		Relocation fields hiding	(empty)		true	true	false	2025-09-18 02:07:31	100	
⏪ ⏩ 1 to 2 of 2 ⏪ ⏩										

b. Creation of Table

i. Creation of Table

1. **Navigate to:** System Definition > **Tables.**

a. Click **New** to create a new table.

b. **Fill in Table Information:**

i. **Name:** Name of the table ----- ii. **Label:** Backend name of the table----- iii.

Auto-generate schema: Leave it checked if you'd like ServiceNow to autogenerate schema fields.

c. Click **Submit** to create the table.

Table Network DataBase Table		Delete	Update	Delete All Records
A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. More Info				
* Label	Network DataBase Table	Application	Global	
* Name	u_network_database_table	Remote Table		

ii. Creation of fields

In ServiceNow, fields are created at the **table** level. To create a field, you first need to identify the table where the field will reside.

1. In the **Application Navigator** (left-side panel), type **Tables** in the search bar.

2. Under **System Definition**, click **Tables**. This will take you to a list of all tables in the

system. **a. Select the Table to Add the Field**

From the list of tables, search for and select the **table** you want to add a field to. For example, if you want to add a field to the **Network database** table:

1. Type "**Network database**" in the search box or scroll through the list.

2. Click on the **Network database** table name. You'll now see a list of all fields

(columns) associated with the **Network database** table.

b. Open the Table's Columns

After selecting the table, you'll be brought to a view that lists all the columns (fields) that currently exist on that table.

To create a new field (column), go to the **Columns** tab (this is where all fields for the selected table are listed).

c. Create a New Field

1. In the **Columns** tab, click the **New** button located at the top-right corner of the page to create a new field.

2. You'll now be prompted with a form where you need to define the new field. The following fields need to be filled out:

Table Columns for text Search						
Dictionary Entries						
	Column label	Type	Reference	Max length	Default value	Display
×	Work Status	String	(empty)	40		false
×	Device Details	String	(empty)	40		false
×	Requested For	String	(empty)	40		false
×	Date of Enquiry	Date	(empty)	40		false
	Sys ID	Sys ID (GUID)	(empty)	32		false
	Created by	String	(empty)	40		false
	Created	Date/Time	(empty)	40		false
	Updated by	String	(empty)	40		false
	Updates	Integer	(empty)	40		false
	Updated	Date/Time	(empty)	40		false
×	Customer Address	String	(empty)	40		false
×	Request Number	String	(empty)	40		false
×	Assignment Group	Reference	Group	32		false
×	Customer Document	String	(empty)	40		false
×	Assigned to	Reference	User	32		false

c. Request Approvals Creation(Related List)

i. Creation of Related List

1. Navigate to **System Definition > Relationships**.
2. Click **New** to create a new relationship.
3. Fill in the following details:
 - o **Name:** Approval Request
 - o **Applies to Table :** Network Database table.
 - o **Queries from Table :** Sysapprovals table.
 - o **Active:** Make sure it's set to **True**.
4. Save the relationship.

Relationship Approval Request		Update	Delete
Name	Approval Request	Application	Global
Advanced	☐	Applies to table	Network DataBase Table [u_network_da...
		Queries from table	Approval [sysapproval_approver]

ii. Adding Related List to the Table

1. Navigate to **Form Designer** for the table where you want to show related records.
2. Add a **Related List** widget to the form.
3. Select the **Related List** you want to show

Query with  Turn on ECMAScript 2021 (ES12) mode 

```
1 (function refineQuery(current, parent) {
2
3     // Add your code here, such as current.addQuery(field, value);
4     current.addQuery('source_table', parent.getTableName());
5     current.addQuery('document_id', parent.sys_id);
6
7 })(current, parent);
```

d. Creation & Implementation of flows, Actions in Flow Designer

i. Creation of Flow

1. Navigate to Flow designer home page
2. Click on New to create a new flow
3. Provide flow name as **Network Request**
4. Provide description of flow 5. Click on Build flow.

ii. Configuring Trigger

1. Click on (+) Icon to Configure the Trigger 2.
- Select Trigger as Application >> Service catalog 3.
- Click on **Done**.

TRIGGER

 Service Catalog  

Trigger

Service Catalog

Advanced Options ▾

Delete

Cancel

Done

iii. Configuring Actions

Click on Actions button to configure new action

1. Get Catalog Variables

- Click on Action, search for Get Catalog Variables
- Select Get Catalog Variables
- Action Inputs>> Trigger>>service catalog>>Requested Item

- Template catalog items >> Select table >> Network Request
- Select the Required Variables and Move to the selected area. • Click on done

1  Get Catalog Variables from Network Request 

Action Properties

Action Get Catalog Variables

Action Inputs

* Submitted Request [Requested Item] Trigger ~... ▶ Requested Item ...  

Select one or more values from the Template Catalog Items and Variable Sets, and select the required Catalog Variables to generate output data pills. You cannot choose the same Catalog Variable from multiple Template Catalog Items and Variable Sets.

* Template Catalog Items and Variable Sets [Catalog Items and Variable Sets] Network Request   

Catalog Variables

Available		Selected
No available values	 	opened_on_behalf_of please_provide_address_here types_of_devices email_id proof_of_document username phonenumber is_this_a_new_connection_or if_this_is_a_relocation_please if_this_is_a_relocation_please provide_device_details

Note: If removing a variable from the 'Selected' list, it will be moved to 'Available' list only if the variable is from the selected Template Catalog Items and Variable Sets. Otherwise, the variable is

2. Create Record

- Select action as Create Record
- Select table as Network Database
- Click on Add fields button to configure the fields
- Configure the Required fields as shown in the below picture
- Click on done

2 Create Network DataBase Table Record

Action Properties

Action: Create Record

Action Inputs

* Table	Network DataBase Table [u_net...	
* Fields	Request Number	Trigger - Servic... ▶ ... ▶ Num...
	Requested For	1 - Get Catalog Va... ▶ ... ▶ Na...
	Work Status	New
	Assignment Group	Network
	Date of Enquiry	Trigger - Service... ▶ ... ▶ Creat...
	Device Details	1 - Get Catal... ▶ types_of_de...
	Customer Address	1 - G... ▶ please_provide_addr...

+ Add field value

Delete Cancel Done

3. Ask for approvals

- Select action as Ask for Approval
- Select target record >> Create record>> network database table
- Provide Approval Reason>> Waiting for approval • Configure approval rules>> Approve, reject, approve/reject
- Select approvals as Anyone approves, everyone approves etc.
- We can select approvals like static/dynamic as shown below • Click on done

3 Ask For Approval

Action Properties

Action: Ask For Approval

Action Inputs

* Record	Trigger -... ▶ Requested Item ...
Table	Requested Item [sc_req_item]
Approval Reason	Waiting for approval
Approval Field	Approval
Journal Field	Approval history

* Rules

Approve When: Anyone approves Network

Due Date: None

Add another OR rule set

OR AND

Delete Cancel Done

4. Flow Logic

- Select action as flow logic and Select If condition
- Apply condition >> Ask for approvals state is **Approved/Rejected** as per requirement
- Click on done

Condition Label: If Request is Approved

* Condition 1: Trigger - Servic... > ... > Appro... is Approved

Add another condition set(OR)

Delete Cancel Done

5. Send Email

- Select action as Send Email
- Select target record >> Create record>> network database table
- Table will be selected automatically
- Configure To, CC, BCC as per our requirements(select static/dynamic)
- Provide Subject & Body as shown in the below picture
- Click on done

then Send Email

Action Properties

Action: Send Email

Action Inputs

Target Record: 2 - ... > Network DataBase Ta...

Table: Network DataBase Table [u_net...

Include Watermark: ☒

* To: 1 - Get Catalog Va... > ... > Em...

CC: 2 - Create Record > ... > Email

BCC:

* Subject: Request had been created

Body

Hello 2 - Create ... > ... > Requested ...

We have received your request with request number: 2 - Creat... > ... > Request Nu...

Sorry for the inconvenience and Your request will be resolved within 2 business working days.

Thanks for contacting us.

Network Team

Delete Cancel Done

6. Update Record

- Select action as Update Record

- Select record as >> create record>> network database
- Table will be selected automatically
- Configure the fields as per requirement, as shown in below
- Click on done

Update Network DataBase Table Record

Action Properties

Action: Update Record

Action Inputs

* Record: 2 - ... > Network DataBase Ta... X

* Table: Network DataBase Table [u_net... X

* Fields:

Assigned to	X	Abel Tuter	X	ⓘ	🔍	⊖
Work Status	X	In Progress		ⓘ	🔍	⊖

+ Add field value

Delete Cancel Done

iv. Flow Chart



Network Request

Active

TRIGGER



Service Catalog

ACTIONS Select multiple

- 1 Get Catalog Variables from Network Request ⓘ
- 2 Create Network DataBase Table Record ⓘ
- 3 Ask For Approval ⓘ
- 4 If If Request is Approved
- 5 then Send Email ⓘ
- 6 Update Network DataBase Table Record ⓘ

6. Screenshots of Output

Home > Service Catalog > Networks and Connectivity > Network Request

Search Catalog

Network Request

Network request Management

* Indicates required

Requester information

Email Id User name

Proof of Document Phone Number

Opened on behalf of

Is this a New connection or Relocation?

☐ New

☐ Relocation

If this is a relocation, Please provide your location

If this is a relocation, Please provide your relocated address here

Please provide address here

* Types of Devices

☐ Laptop

☐ Mobile

☐ Others

☒ None

* Provide device details here

If anything else, please specify

[Add attachments](#)

Quantity:

Delivery Time: 0 Days

Required information

[Provide device details here](#)

fig : Creation of Service Catalog

EXECUTION DETAILS Network Request					
Show Action Details		Test Run - Completed	Open flow	Open context record	
FLOW STATISTICS		Run as: System Administrator	Open flow logs	Completed	2025-09-18 08:04:11
TRIGGER		Catalog Item Requested			
ACTIONS					
1	Get Catalog Variables from Network Request	Core Action	Completed	2025-09-18 08:04:11	24ms
2	Create Record	Core Action	Completed	2025-09-18 08:04:11	10ms
3	Ask For Approval	Core Action	Completed	2025-09-18 08:04:11	504ms
4	If Request is Approved	Flow Logic	Evaluated - True	2025-09-18 08:04:11	18ms
5	Send Email		Completed	2025-09-18 08:04:11	18ms
6	Update Record	Core Action	Completed	2025-09-18 08:04:11	3ms

fig : Test for flow execution

Preview Email



Close

Hello Abraham Lincoln

We have received your request with request number: REQ0010006,

Sorry for the inconvenience and Your request will be resolved within 2 business working days.

Thanks for contacting us.

Network Team

Ref:MSG0000329_NOe0ChMxVT9VrTxpkI5l

fig : Email Notification

7. Conclusion

The **Automated Network Request Management in ServiceNow** project provides a simple and efficient way to manage network-related requests.

The project uses **Service Catalog, Variables, Catalog UI Policies, Tables, Approval processes, and Flow Designer** to automate the request process.

Instead of handling every request manually, ServiceNow can automatically create records, process approvals, send notifications, and update request information.

As a result, the solution helps **save time, reduce manual effort, improve request tracking, and provide a better experience for both users and IT teams.**